

## Technology Stack

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| <b>Date</b>          | 26 June 2026                                        |
| <b>Team ID</b>       | SWTID-2026-9014                                     |
| <b>Project Name</b>  | AdForge AI: AI-Powered Advertisement Copy Generator |
| <b>Maximum Marks</b> | 2 Marks                                             |

### Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

### Technology Stack Details

| S.No | Architecture Component / Layer | Technology Chosen                           | Justification / Purpose                                                                                                                         |
|------|--------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Frontend / Client-Side         | HTML5,<br>CSS3,<br>JavaScript,<br>Marked.js | Used to build a responsive and interactive user interface, handle user input, and display AI-generated advertisement content in a clean format. |

|   |                                     |                                                |                                                                                                                                  |
|---|-------------------------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 2 | <b>Backend / Server-Side</b>        | <i>Python 3.8+, Flask 3.0.3</i>                | <i>Flask handles routing, request processing, API integration, and business logic for AI-powered advertisement generation.</i>   |
| 3 | <b>Database / Data Storage</b>      | <i>No Database (Stateless Application)</i>     | <i>The application processes requests in real time using the Groq API and does not permanently store user data.</i>              |
| 4 | <b>Cloud / Hosting / Deployment</b> | <i>Ngrok (Development), Flask Local Server</i> | <i>Ngrok exposes the local Flask application through a secure public URL for testing, demonstrations, and mentor evaluation.</i> |
| 5 | <b>Version Control &amp; CI/CD</b>  | <i>Git &amp; GitHub</i>                        | <i>Git is used for version control, while GitHub enables team collaboration, code management, and project hosting.</i>           |

|          |                                           |                                                                                                  |                                                                                                                                                                        |
|----------|-------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |                                           |                                                                                                  |                                                                                                                                                                        |
| <b>6</b> | <b>Third-Party APIs /<br/>Other Tools</b> | <i>Groq Cloud<br/>API,<br/>Qwen/Qwen<br/>3-32B<br/>Model,<br/>python-<br/>dotenv,<br/>Bleach</i> | <i>Groq API generates AI<br/>advertisement content,<br/>python-dotenv securely<br/>manages API keys, and<br/>Bleach sanitizes user inputs<br/>to improve security.</i> |